

COOLANGATTA, Australia

WMO: 945920

Lat: 28.168S

Lon: 153.505E

Elev: 15

StdP: 14.69

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 42.8	(c) 46.0	(d) 31.0	(e) 25.2	(f) 61.9	(g) 34.7	(h) 29.4	(i) 58.8	(j) 21.0	(k) 64.8	(l) 19.1	(m) 65.3	(n) 4.0	(o) 230	(p) 0.425

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 11.3	(c) 85.5	(d) 73.9	(e) 84.0	(f) 73.9	(g) 82.5	(h) 72.9	(i) 77.4	(j) 82.1	(k) 76.2	(l) 81.0	(m) 75.1	(n) 80.1	(o) 15.3	(p) 10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.8	134.9	80.2	74.7	130.2	79.6	73.5	124.6	78.7	40.9	82.3	39.7	80.9	38.7	80.4	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 22.0	(b) 20.2	(c) 18.4	(e) 37.7	(f) 92.3	(g) 2.1	(h) 3.7	(i) 36.2	(j) 94.9	(k) 35.0	(l) 97.0	(m) 33.8	(n) 99.1	(o) 32.2	(p) 101.7		
(a) 22.0	(b) 20.2	(c) 18.4	(e) 36.0	(f) 79.1	(g) 2.0	(h) 1.1	(i) 34.6	(j) 79.9	(k) 33.4	(l) 80.5	(m) 32.3	(n) 81.1	(o) 30.9	(p) 81.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	68.3	76.2	75.8	74.0	69.9	65.0	61.2	59.5	60.7	64.8	68.1	71.3	74.0
	DBStd	6.71	2.71	2.88	2.64	2.85	3.57	3.84	3.69	3.82	3.69	3.54	3.58	3.17
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	543	0	0	0	3	43	121	174	142	48	11	1	0
	CDD50	6692	811	724	745	598	465	335	294	332	444	560	640	744
	CDD65	1759	346	304	280	150	43	6	4	9	41	106	191	279
	CDH74	9139	2392	2022	1484	486	80	5	7	29	111	280	743	1500
	CDH80	1146	370	336	154	19	0	0	1	2	15	28	64	157
(14) Wind	WSAvg	9.5	10.6	10.3	9.7	8.9	7.9	8.0	7.9	8.5	9.7	10.6	10.7	10.6
(15) Precipitation	PrecAvg	67.3	7.2	8.1	9.5	7.0	6.5	4.4	3.5	2.3	2.0	4.8	5.2	6.1
	PrecMax	110.6	20.3	20.2	29.2	28.7	20.0	25.0	15.5	6.6	9.1	25.2	15.7	18.2
	PrecMin	37.9	0.6	0.6	0.9	1.5	0.1	0.2	0.0	0.1	0.0	0.0	0.1	0.9
	PrecStd	20.6	4.8	4.7	7.7	5.8	4.7	5.2	3.5	1.7	1.9	5.2	3.7	4.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.8	86.9	85.7	82.2	78.5	74.6	74.2	77.5	81.2	83.8	85.7	86.4
		MCWB	76.1	75.9	75.1	70.3	65.6	63.4	60.9	61.2	61.5	64.7	67.8	70.9
	2%	DB	84.6	84.7	82.8	80.1	76.2	72.3	71.7	73.5	76.7	78.7	81.0	82.8
		MCWB	74.8	75.0	73.2	69.2	65.2	62.6	60.8	61.3	64.1	67.7	71.1	72.9
	5%	DB	83.5	83.7	81.8	78.5	74.5	70.8	70.0	71.5	74.4	76.8	79.3	81.6
		MCWB	74.1	74.4	72.4	68.5	64.5	61.5	60.2	60.8	63.9	67.4	70.3	72.3
	10%	DB	82.2	82.2	80.4	76.9	73.0	69.1	68.2	69.7	72.8	75.2	78.0	80.2
		MCWB	73.3	73.1	71.3	67.8	63.8	60.4	58.9	59.8	63.2	66.3	69.5	71.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.1	79.0	77.4	73.6	71.5	67.3	67.1	67.7	70.0	72.2	75.9	77.0
		MCDB	83.3	83.8	81.9	78.5	73.9	70.9	71.0	72.7	73.4	76.4	79.0	81.8
	2%	WB	77.4	77.5	75.8	71.6	69.0	65.4	64.6	65.3	68.1	70.6	73.8	75.5
		MCDB	82.0	82.6	80.5	77.1	73.3	69.5	69.2	70.3	73.0	74.9	78.2	80.4
	5%	WB	76.1	76.2	74.4	70.3	66.9	63.8	62.4	63.4	66.6	69.4	72.2	74.3
		MCDB	80.9	81.3	79.5	76.0	71.8	68.4	67.3	68.9	71.9	74.2	77.3	79.2
	10%	WB	74.8	75.0	73.1	69.0	65.3	62.4	60.5	61.7	65.3	68.2	70.9	73.2
		MCDB	80.0	80.1	78.4	74.9	70.5	67.1	66.1	67.6	70.8	73.6	76.2	78.1
(35) Mean Daily Temperature Range	5% DB	MDBR	11.3	11.7	12.2	13.6	15.5	15.5	17.7	18.2	16.2	13.8	12.4	11.9
		MCDBR	12.7	13.0	13.7	15.7	17.5	17.3	20.0	20.8	18.5	15.9	13.4	13.5
		MCWBR	6.1	6.1	6.4	7.8	8.9	9.4	11.1	11.4	10.1	8.9	6.9	6.7
	5% WB	MCDBR	10.7	11.0	11.7	13.8	13.1	13.5	15.8	16.6	14.1	13.0	11.6	11.3
		MCWBR	6.1	6.1	6.5	7.7	7.6	8.3	10.3	10.4	9.0	8.3	6.6	6.4
(40) Clear-Sky Solar Irradiance	taub		0.391	0.377	0.360	0.341	0.321	0.314	0.305	0.318	0.360	0.387	0.384	0.397
	taud		2.500	2.553	2.597	2.627	2.635	2.649	2.657	2.605	2.479	2.431	2.482	2.469
	Ebn at Noon		302	301	296	287	279	274	282	291	291	294	302	301
	Edh at Noon		36	34	30	27	24	23	23	27	34	38	37	38
(44) All-Sky Solar Radiation	RadAvg		2046	1863	1593	1342	1116	922	1076	1357	1679	1876	2046	2037
	RadStd		163	149	90	67	63	62	85	105	134	162	159	212

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+0.81	N/A	-0.81	N/A	N/A	+212	+162
(47) Regional (25 neighbors)	+0.72	N/A	N/A	N/A	N/A	N/A	N/A	-115	+328	+198

Nomenclature: See separate page